

Structural Break and Time Series Modeling of Weekly Cereal Sales

Alex Spigler

April 4, 2025

Introduction

This report analyzes weekly cereal sales for a brand over a two-year period. At week 88, a competitor introduced a similar product into the market. The goal is to evaluate whether this competitive entry resulted in a structural change in sales patterns. Understanding structural breaks is important for businesses to identify the impact of interventions such as marketing campaigns, pricing changes, supply chain disruptions, or competitive actions.

The goals of this analysis are to:

1. Model the underlying trend and a possible level/trend shift at the intervention week.
2. Diagnose any violations of classical regression assumptions, especially autocorrelation.
3. Produce short-term forecasts with prediction intervals.

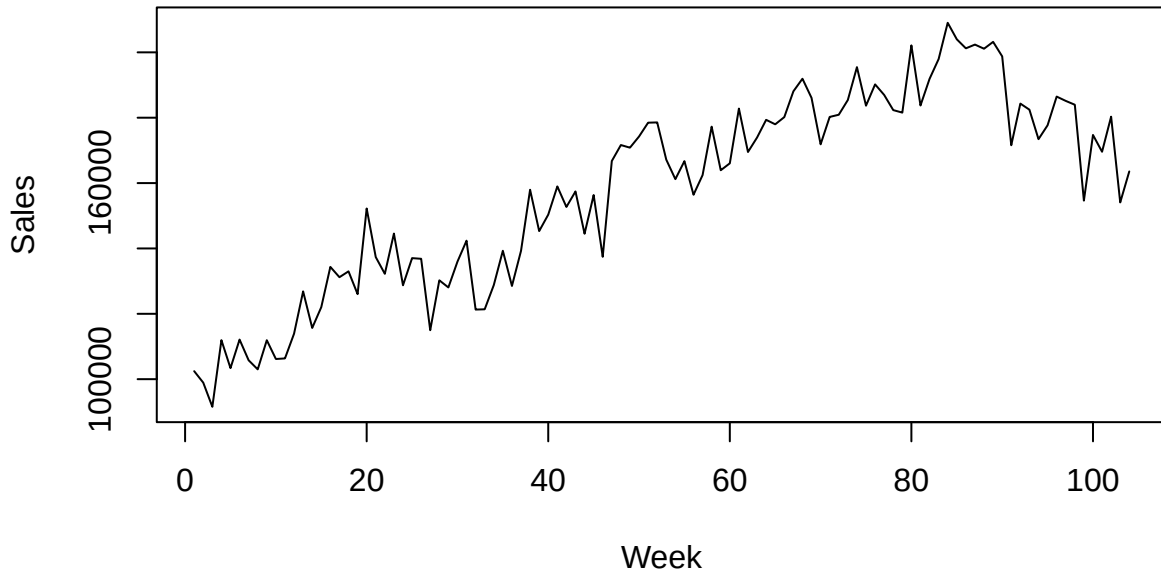
The analysis is carried out in **R** using a regression model with an intervention term and ARIMA errors.

Data and Exploratory Analysis

```
cereal <- read.csv("cereal-sales-data.csv")
cereal_ts <- ts(cereal$sales, start = 1, frequency = 52)

# Plot with week numbers on x-axis
plot(
  x = 1:length(cereal_ts),
  y = as.numeric(cereal_ts),
  type = "l",
  main = "Weekly Cereal Sales",
  xlab = "Week",
  ylab = "Sales"
)
```

Weekly Cereal Sales



Visually, the series shows a non-constant mean and an upward trend, indicating nonstationarity. There also appears to be a noticeable change around week 88, motivating an intervention model.

Methods

Regression with Intervention

An intervention regression model is appropriate for modeling a structural break at a known intervention point. Let:

- y_t = cereal sales in week t
- $x_t = 0$ for $t < 88$, $x_t = 1$ for $t \geq 88$
- t = week index
- $x_t t$ = interaction term

The model is:

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 t + \beta_3 x_t t + \varepsilon_t$$

This specification is appropriate because:

- β_0 captures the intercept (baseline level of sales) before week 88
- $\beta_2 t$ captures the trend (rate of change in sales) before week 88
- $\beta_1 x_t$ allows for a jump or drop in level at week 88
- $\beta_3 x_t t$ allows for a change in trend after week 88 (sales could start declining faster or growing more slowly)

```
n <- nrow(cereal)
cereal$t <- seq_len(n)
cereal$x <- ifelse(cereal$t < 88, 0, 1)
```

```

cereal$xt <- cereal$x * cereal$t

reg_model <- lm(sales ~ x + t + xt, data = cereal)
summary(reg_model)

##
## Call:
## lm(formula = sales ~ x + t + xt, data = cereal)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19259.9  -5478.9   -187.6    7303.8   26571.4
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 102585.02    2001.72  51.248 < 2e-16 ***
## x           278482.63    44089.18   6.316 7.46e-09 ***
## t           1153.18      39.51   29.186 < 2e-16 ***
## xt          -3256.83     459.89  -7.082 2.02e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9255 on 100 degrees of freedom
## Multiple R-squared:  0.9077, Adjusted R-squared:  0.9049
## F-statistic: 327.8 on 3 and 100 DF,  p-value: < 2.2e-16

```

Interpretation of coefficients:

All four coefficients are statistically significant ($p < 0.001$). Because the model includes the trend-interaction term $x_t t$, the step and interaction coefficients are *not* interpretable on their own; the immediate jump and the post-intervention slope each combine two coefficients.

- $\hat{\beta}_0 = 102,585$: Baseline sales level before week 88
- $\hat{\beta}_2 = 1,153$: Pre-intervention trend (sales increased by approximately 1,153 units per week)
- $\hat{\beta}_1 = 278,483$: Coefficient on the step dummy x_t (the gap between the two fitted lines extrapolated back to $t = 0$), **not** the jump at week 88
- $\hat{\beta}_3 = -3,257$: Coefficient on the interaction $x_t t$, i.e. the *change* in slope at week 88, **not** the post-intervention slope itself

Combining the relevant terms gives the quantities of interest:

- **Immediate level shift at week 88** $= \hat{\beta}_1 + \hat{\beta}_3 \cdot 88 = 278,483 - 286,616 \approx -8,100$ units. The intervention produces little immediate jump; the raw series is essentially flat across the break (weeks 87 to 89: 202,370, 201,100, 203,210).
- **Post-intervention trend** $= \hat{\beta}_2 + \hat{\beta}_3 = 1,153 - 3,257 = -2,104$ units/week. Competitor entry reverses the trend from +1,153 units/week (growth) to -2,104 units/week (decline), a change in slope of -3,257 units/week.

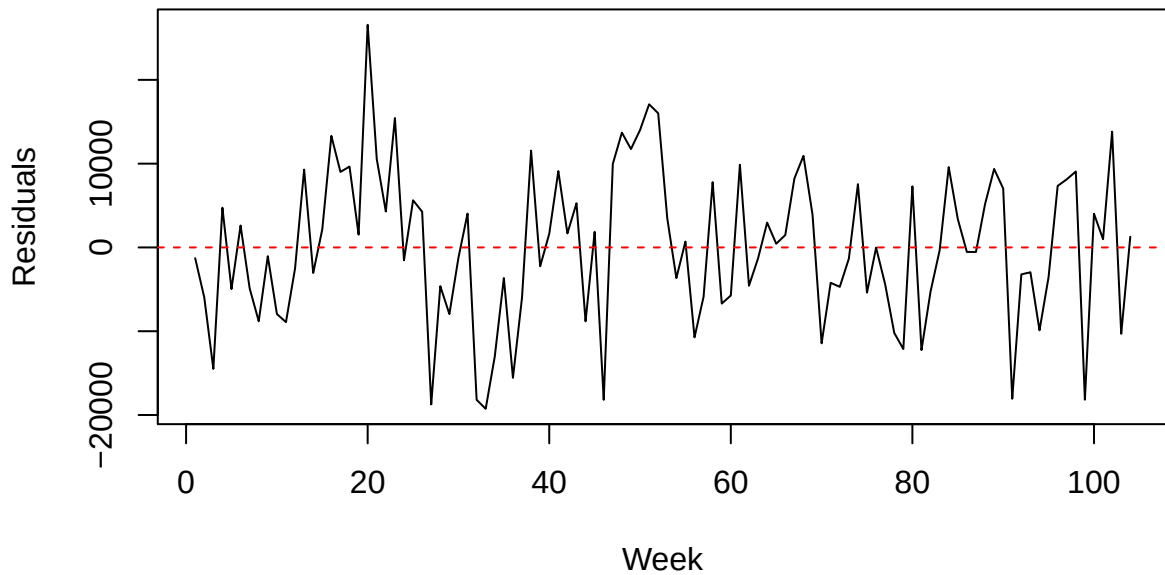
Regression Diagnostics

```
reg_resid <- resid(reg_model)
par(mfrow = c(2, 1))

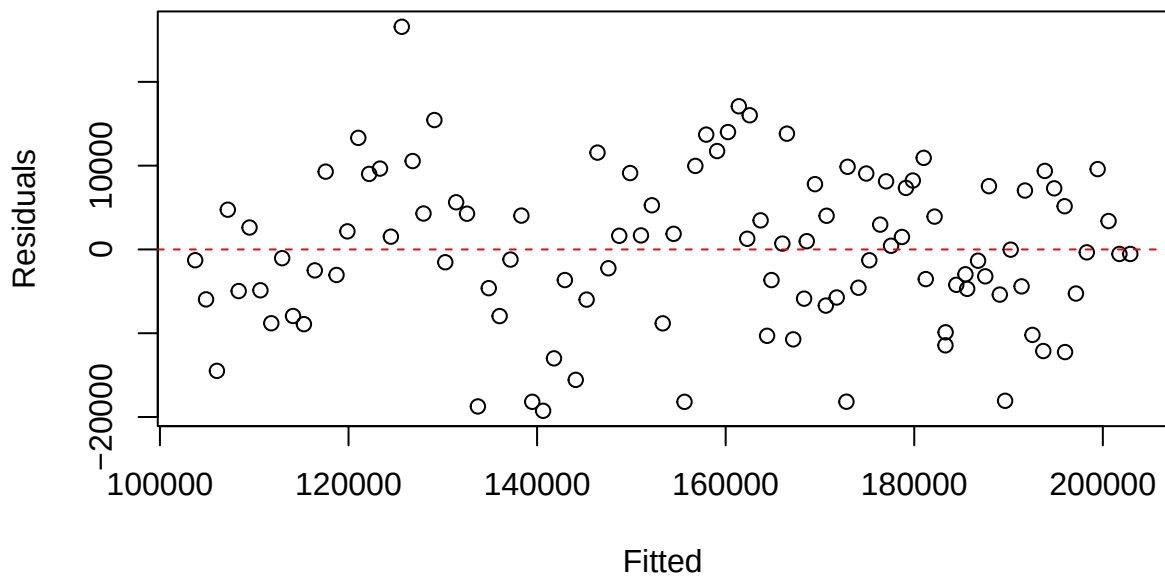
plot(reg_resid, type="l", main="Residuals vs Time", xlab="Week", ylab="Residuals")
abline(h=0, col="red", lty=2)

plot(fitted(reg_model), reg_resid, main="Residuals vs Fitted", xlab="Fitted", ylab="Residuals")
abline(h=0, col="red", lty=2)
```

Residuals vs Time



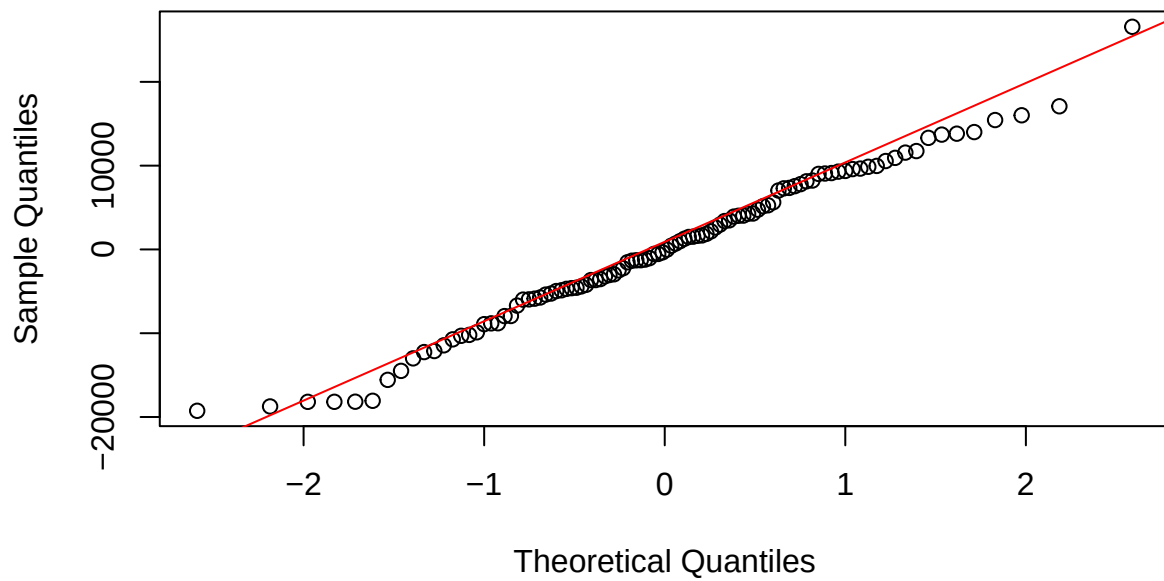
Residuals vs Fitted



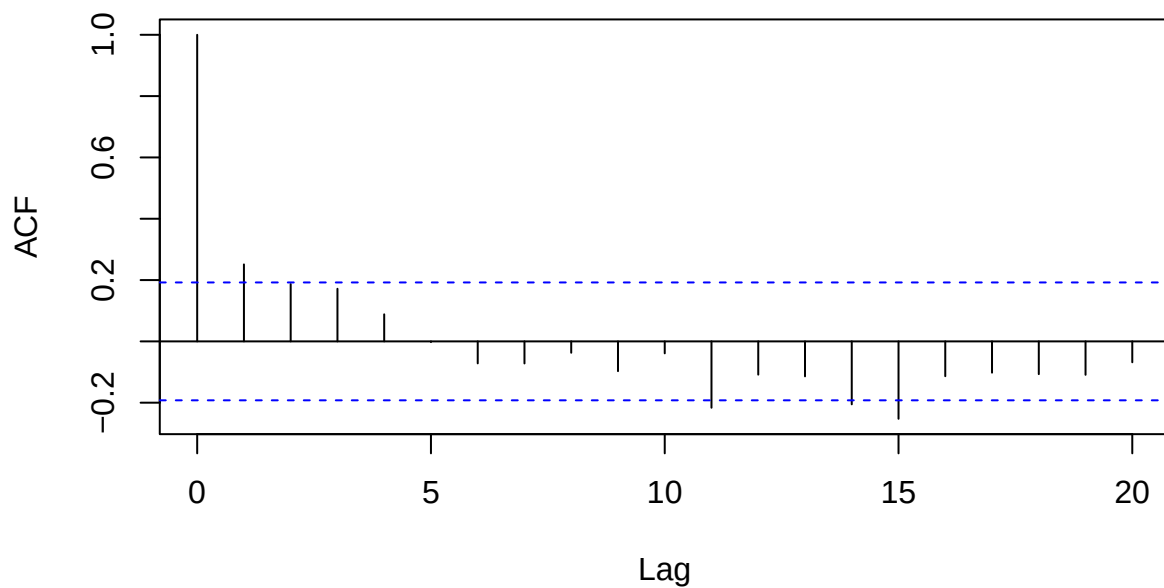
```
qqnorm(reg_resid, main="Normal Q-Q Plot")
qqline(reg_resid, col="red")

acf(reg_resid, main="ACF of Regression Residuals")
```

Normal Q-Q Plot



ACF of Regression Residuals



```
par(mfrow = c(1, 1))
```

Tests

```
dwtest(reg_model)
```

```
##
```

```
## Durbin-Watson test
##
## data: reg_model
## DW = 1.4974, p-value = 0.001955
## alternative hypothesis: true autocorrelation is greater than 0
```

```
ncvTest(reg_model)
```

```
## Non-constant Variance Score Test
## Variance formula: ~ fitted.values
## Chisquare = 1.114541, Df = 1, p = 0.2911
```

```
shapiro.test(reg_resid)
```

```
##
## Shapiro-Wilk normality test
##
## data: reg_resid
## W = 0.98846, p-value = 0.5139
```

Independence: While the residuals vs. time plot does not show an obvious visual pattern, formal testing reveals positive autocorrelation. The Durbin-Watson test (p-value = 0.001955) rejects the null hypothesis of no autocorrelation. This is further confirmed by the ACF plot, which shows a significant spike at lag 1.

Constant Variance: The residuals vs. fitted values plot shows residuals randomly distributed around $y = 0$, suggesting constant variance. This is supported by the Cook-Weisberg non-constant variance score test (p-value = 0.2911), which checks whether the residual variance scales with the fitted level — the natural heteroscedasticity concern for a trending series — and does not reject the null hypothesis of constant variance.

Normality: The Q-Q plot appears approximately normal, and the Shapiro-Wilk test (p-value = 0.5139) does not reject the null hypothesis of normality.

Conclusion: The only assumption that fails is independence, due to the presence of positive autocorrelation among the error terms. This motivates the use of ARIMA errors.

Model Refinement with ARIMA Errors

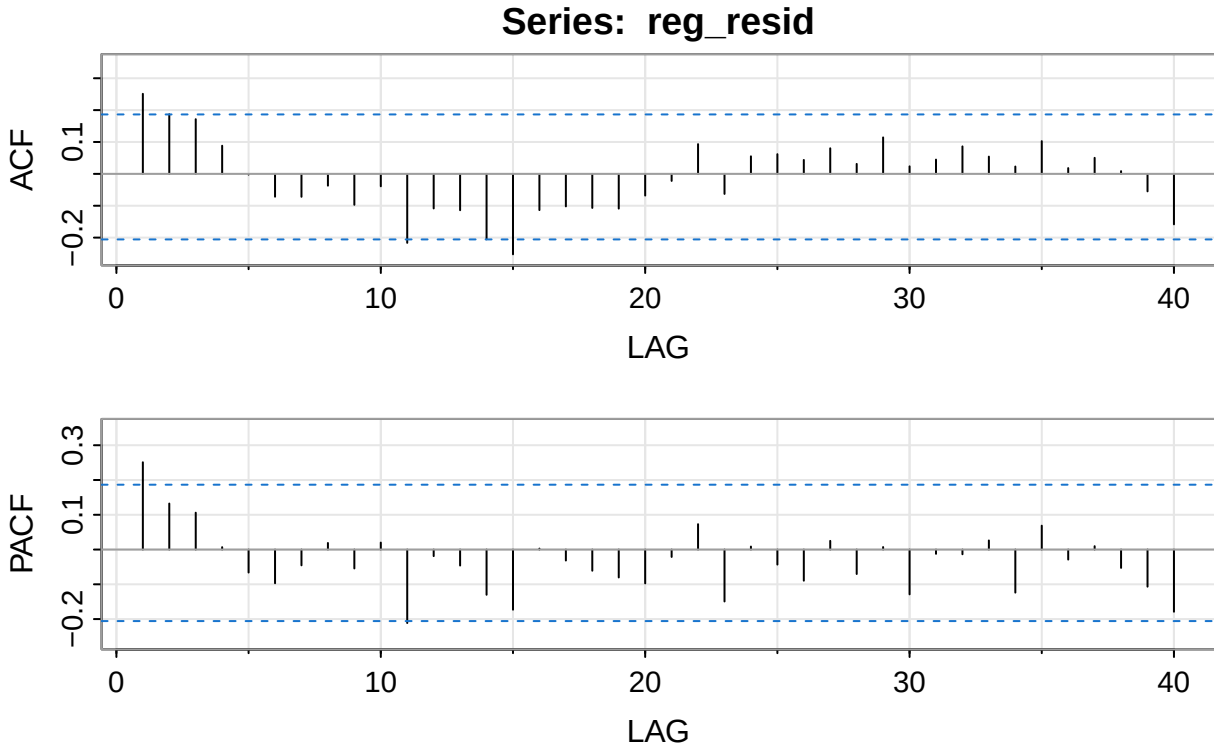
Residual Analysis

Because the diagnostic tests showed positive autocorrelation among the error terms, the residuals are not white noise and require further modeling.

```
kpss.test(reg_resid)
```

```
##
## KPSS Test for Level Stationarity
##
## data: reg_resid
## KPSS Level = 0.047805, Truncation lag parameter = 4, p-value = 0.1
```

```
acf2(reg_resid, max.lag = 40)
```



```
##      [,1] [,2] [,3] [,4]  [,5]  [,6]  [,7]  [,8]  [,9] [,10] [,11] [,12] [,13]
## ACF  0.25 0.19 0.17 0.09  0.00 -0.07 -0.07 -0.04 -0.10 -0.04 -0.22 -0.11 -0.11
## PACF 0.25 0.13 0.11 0.01 -0.07 -0.10 -0.05  0.02 -0.05  0.02 -0.21 -0.02 -0.05
##      [,14] [,15] [,16] [,17] [,18] [,19] [,20] [,21] [,22] [,23] [,24] [,25]
## ACF  -0.21 -0.25 -0.11 -0.10 -0.11 -0.11 -0.07 -0.02  0.09 -0.06  0.06  0.06
## PACF -0.13 -0.17  0.00 -0.03 -0.06 -0.08 -0.10 -0.02  0.07 -0.15  0.01 -0.04
##      [,26] [,27] [,28] [,29] [,30] [,31] [,32] [,33] [,34] [,35] [,36] [,37]
## ACF   0.04  0.08  0.03  0.11  0.02  0.04  0.09  0.05  0.02  0.10  0.02  0.05
## PACF -0.09  0.02 -0.07  0.01 -0.13 -0.01 -0.01  0.03 -0.12  0.07 -0.03  0.01
##      [,38] [,39] [,40]
## ACF   0.01 -0.05 -0.16
## PACF -0.05 -0.11 -0.18
```

Stationarity Check: The KPSS test (p-value > 0.1) does not reject the null hypothesis of level stationarity, confirming the residuals have constant mean and variance. Therefore, differencing is not needed ($d = 0$ in ARIMA notation).

Seasonality Check: The ACF shows no significant spikes at seasonal lags such as 52 or 104, indicating no seasonal autocorrelation. Seasonal differencing is not appropriate for this series.

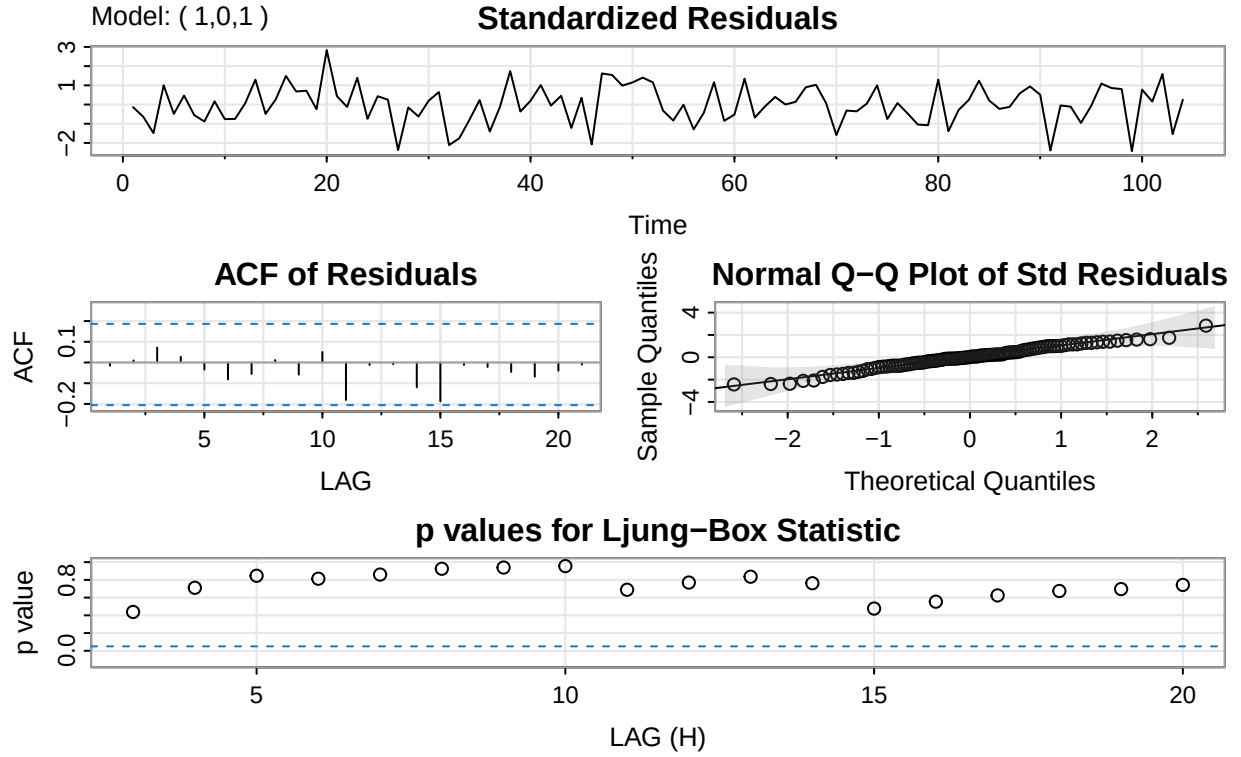
Model Selection: The ACF shows a significant spike at lag 1 that gradually tails off. The PACF similarly shows a significant spike at lag 1 that tails off. This pattern (both ACF and PACF showing initial spikes that decay) suggests an ARIMA(1,0,1) model is appropriate for the residuals. A simpler AR(1) model fits the residuals nearly as well, but both the AR and MA terms are statistically significant and the gradual tail-off in both the ACF and PACF supports the mixed ARMA(1,1) specification, so the MA(1) term is retained.

ARIMA(1,0,1) Error Model

$$e_t = \phi_1 e_{t-1} + \varepsilon_t + \theta_1 \varepsilon_{t-1}$$

```
sarima(reg_resid, 1, 0, 1)
```

```
## initial value 9.118047
## iter 2 value 9.098328
## iter 3 value 9.088789
## iter 4 value 9.088293
## iter 5 value 9.087804
## iter 6 value 9.081453
## iter 7 value 9.075473
## iter 8 value 9.074289
## iter 9 value 9.073717
## iter 10 value 9.073354
## iter 11 value 9.073352
## iter 11 value 9.073352
## final value 9.073352
## converged
## initial value 9.069797
## iter 2 value 9.069757
## iter 3 value 9.069732
## iter 4 value 9.069729
## iter 5 value 9.069728
## iter 6 value 9.069725
## iter 7 value 9.069724
## iter 8 value 9.069723
## iter 8 value 9.069723
## final value 9.069723
## converged
## <><><><><><><><><><><>
##
## Coefficients:
## Estimate SE t.value p.value
## ar1 0.6728 0.1747 3.8513 0.0002
## ma1 -0.4514 0.2034 -2.2200 0.0287
## xmean -45.5615 1411.7357 -0.0323 0.9743
##
## sigma^2 estimated as 75407844 on 101 degrees of freedom
##
## AIC = 21.05425 AICc = 21.05655 BIC = 21.15595
##
```



Model Diagnostics:

- The standardized residuals show no structure and vary randomly around 0
- The ACF of residuals is within bounds, indicating white noise
- The Q-Q plot appears normal
- The p-values for the Ljung-Box statistic are all above 0.05, indicating no remaining autocorrelation
- Both AR(1) and MA(1) terms are statistically significant (p-values of 0.0002 and 0.0287 respectively)
- AIC = 21.05425, BIC = 21.15595

This is a well-fitting model. Residuals now appear to be white noise, validating the combined regression + ARIMA(1,0,1) approach.

Combined Model Specification

The final model has a regression component with an intervention at Week 88, and ARIMA(1,0,1) errors to account for autocorrelation in the residuals:

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 t + \beta_3 x_t t + e_t$$

where the error terms e_t follow an ARIMA(1,0,1) process:

$$e_t = \phi_1 e_{t-1} + \varepsilon_t + \theta_1 \varepsilon_{t-1}$$

with:

- ε_t = white noise
- $\phi_1 = 0.6728$ = autoregressive parameter
- $\theta_1 = -0.4514$ = moving average parameter

This model allows for a structural break in both the level and trend of sales at Week 88, and it captures autocorrelated, stationary errors. As a robustness check, re-estimating the regression coefficients jointly with the ARIMA(1,0,1) errors (generalized least squares rather than OLS) yields a very similar post-intervention slope, confirming the trend reversal is a genuine feature of the data and not an artifact of the OLS fit, whose standard errors the autocorrelation had invalidated.

*# GLS check: re-fit the regression jointly with the ARIMA(1,0,1) errors and
compare the post-intervention slope to the OLS estimate.*

```
gls_fit <- arima(
  cereal_ts,
  order = c(1, 0, 1),
  xreg = cbind(x = cereal$x, t = cereal$t, xt = cereal$xt)
)
ols_slope <- coef(reg_model)["t"] + coef(reg_model)["xt"]
gls_slope <- coef(gls_fit)["t"] + coef(gls_fit)["xt"]
cat(sprintf("OLS post-intervention slope: %.1f units/week\n", ols_slope))
```

```
## OLS post-intervention slope: -2103.7 units/week
```

```
cat(sprintf("GLS post-intervention slope: %.1f units/week\n", gls_slope))
```

```
## GLS post-intervention slope: -2236.8 units/week
```

Forecasting

```
xreg <- cbind(cereal$x, cereal$t, cereal$xt)

h <- 10
future_t <- (n + 1):(n + h)
future_x <- ifelse(future_t < 88, 0, 1)
future_xt <- future_x * future_t
newxreg <- cbind(future_x, future_t, future_xt)

fc <- sarima.for(
  x      = cereal_ts,
  n.ahead = h,
  p      = 1,
  d      = 0,
  q      = 1,
  xreg    = xreg,
  newxreg = newxreg,
  plot    = FALSE
)

forecast_df <- data.frame(
```

```

Week      = future_t,
Forecast  = round(fc$pred, 2),
Lower95   = round(fc$pred - 1.96 * fc$se, 2),
Upper95   = round(fc$pred + 1.96 * fc$se, 2)
)
forecast_df

```

```

##      Week Forecast  Lower95  Upper95
## 1    105 159470.0 142456.0 176483.9
## 2    106 157193.5 139761.0 174626.1
## 3    107 154929.9 137309.5 172550.3
## 4    108 152675.0 134969.4 170380.6
## 5    109 150426.0 132681.7 168170.3
## 6    110 148180.9 130418.9 165942.9
## 7    111 145938.6 128168.5 163708.6
## 8    112 143698.0 125924.3 161471.7
## 9    113 141458.7 123683.3 159234.1
## 10   114 139220.2 121444.0 156996.4

```

```

# Base plot: historical data

```

```

plot(
  x = 1:length(cereal_ts),
  y = as.numeric(cereal_ts),
  type = "l",
  xlim = c(1, n + h),
  ylim = range(
    c(
      as.numeric(cereal_ts),
      fc$pred + 1.96 * fc$se,
      fc$pred - 1.96 * fc$se
    )
  ),
  main = "Cereal Sales: Historical Data and 10-Week Forecast",
  xlab = "Week",
  ylab = "Sales"
)

```

```

# Future x-values (weeks)

```

```

future_index <- (n + 1):(n + h)

```

```

# Forecast line

```

```

lines(future_index, fc$pred, col = "blue", lwd = 2)

```

```

# 95% prediction intervals

```

```

lines(future_index, fc$pred + 1.96 * fc$se, col = "blue", lty = 2)
lines(future_index, fc$pred - 1.96 * fc$se, col = "blue", lty = 2)

```

```

legend(

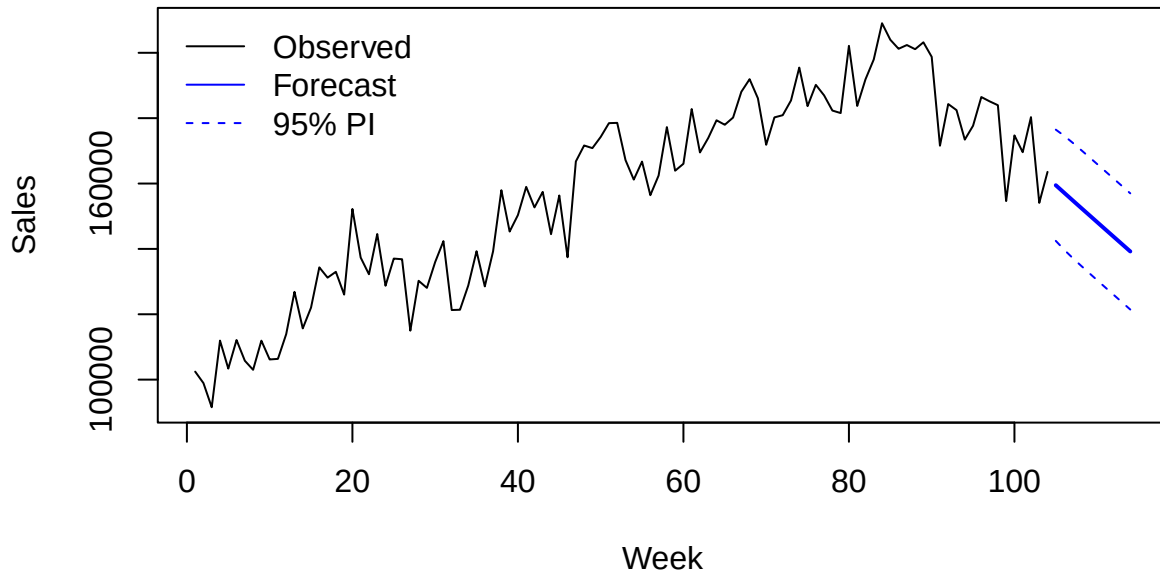
```

```

"topleft",
legend = c("Observed", "Forecast", "95% PI"),
col     = c("black", "blue", "blue"),
lty     = c(1, 1, 2),
bty     = "n"
)

```

Cereal Sales: Historical Data and 10-Week Forecast



Conclusion

This analysis successfully modeled weekly cereal sales with a structural break at week 88 using a combined regression and ARIMA approach.

Key Findings:

- Regression analysis revealed a significant structural break at week 88, coinciding with the competitor's market entry. The effect is almost entirely a change in trend, not an immediate jump: the level shift at week 88 is small (about -8,100 units, with the raw series flat across the break), while the weekly trend reverses from +1,153 units/week (growth) to -2,104 units/week (decline), a change in slope of -3,257 units/week consistent with sustained competitive pressure.
- Standard regression residuals exhibited significant positive autocorrelation, violating the independence assumption.
- An ARIMA(1,0,1) error structure effectively captured the autocorrelation, with all diagnostic tests confirming the residuals now behave as white noise.
- The combined model fits the data well and produces stable 10-week forecasts with reasonable prediction intervals.

Potential Extensions:

Future work could explore:

- Analysis of competitor pricing, marketing spend, or product features to explain the magnitude of the sales impact
- Incorporating additional explanatory variables (such as promotions or other competitor actions)
- Extending the forecast horizon and validating predictions against actual data